

NGUYEN MAI HONG TRAM

AI Researcher

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Research Profile

My research has evolved from computer vision and medical visual question answering toward multimodal and video understanding, with recent work on temporally grounded action understanding in egocentric video. I am particularly interested in how vision-language models can move beyond static perception to reason about actions, time, and embodied interaction.

Education

B.Sc. in Data Science Sep 2022 – Mar 2026
University of Economics Ho Chi Minh City (UEH) GPA: 3.92/4.00

- Rank 2 in Data Science; Top 5% of the School of Business Information Technology.
- Merit Scholarships for outstanding academic and extracurricular performance, 2023 and 2025.
- Selected coursework: Data Structures & Algorithms (9.8/10), Mathematics for Computer Science (9.8/10), and Artificial Intelligence (10.0/10).

Selected Programs

Applied AI Talent Program — VinGroup × VinUniversity May 2026 – Aug 2026
Intensive AI training, applied capstone, and industry research internship at VinRobotics

AIO2026 — AI VIETNAM Jun 2026 – Present
Research-oriented artificial intelligence training and community activities

Research & Industry Experience

AI Research Intern Jul 2026 – Aug 2026
VinRobotics — AI Platform, VLA Team Hanoi, Vietnam

- Developed and evaluated multi-stage vision-language pipelines for temporally grounded action captioning on egocentric video.
- Designed controlled experiments on frame sampling, temporal grounding, prompting, and model adaptation; analyzed failure modes including directional ambiguity, dataset shift, and overfitting.
- Investigated recent VLM/VLA methods through literature review, implementation analysis, and large-scale experimentation using SGLang.

AI Research Engineer Intern Mar 2026 – May 2026
TMA Solutions Ho Chi Minh City, Vietnam

- Investigated deep-learning approaches for bidirectional sign-language communication through literature review, algorithm analysis, implementation, and experimental evaluation.
- Researched continuous sign-language recognition and text-to-3D-avatar generation, achieving approximately 5% sequence prediction error on the test set and 10% in a real-time video-chat setting.
- Translated research prototypes into a browser inference pipeline using PyTorch → ONNX Runtime Web, reducing inference time by approximately 37%.

Research Collaborator Mar 2025 – Jan 2026
University of Economics Ho Chi Minh City Ho Chi Minh City, Vietnam

- Conducted research across surgical VQA, Vietnamese scene-text recognition, and multi-objective optimization.
- Contributed throughout the research lifecycle: problem formulation, literature review, data preparation, model development, experimental design, evaluation, and manuscript preparation.

Publications

GNN-SurgVQA: Object-Centric Graph Reasoning for Visual Question Answering in Laparoscopic Scene Understanding International Conference on Data Analytics and Management (ICDAM), 2026 — Best Paper · [Code]
Object-centric multimodal reasoning for surgical VQA; evaluated on 960K+ QA pairs, achieving 85.9% accuracy and 85.8% F1.

Comprehensive Approach to Vietnamese Scene Text Recognition: Challenges, Models, and Framework Development ICDAM 2025, Lecture Notes in Networks and Systems, vol. 1601, Springer, 2025 · [Paper]
End-to-end Vietnamese scene-text detection and recognition; fine-tuning on approximately 25K samples improved recognition performance by more than 2%.

Selected Projects

Multi-Objective Optimization for Emergency Blood Transportation [Code]
Research Project · UEH Young Researchers Award — Prize B · Eureka 2025 Semifinalist

- Formulated emergency blood transportation as a multi-objective EVRP with battery-swapping stations and time windows, jointly optimizing logistics cost and hospital waiting time.
- Developed hybrid NSGA-II/SPEA2 methods with Savings, 2-opt, and 3-opt heuristics.
- Validated on standard EVRP-TW benchmarks and a real Ho Chi Minh City case study involving 36 hospitals and 6 battery-swapping stations.

GradeMind — AI-Assisted Mathematics Grading Platform [Demo]
Capstone Project · VinGroup × VinUniversity Applied AI Talent Program

- Built a teacher-in-the-loop AI system for grading Vietnamese secondary-school mathematics with evidence-backed feedback and class-level error analysis.
- Developed a multimodal grading and RAG pipeline with confidence-based routing, source-grounded evidence, PII safeguards, and golden-set evaluation.
- Stack: FastAPI · PostgreSQL/pgvector · Celery · Redis · React/TypeScript · Docker.

Technical Skills

Research	Multimodal Learning, Vision–Language Models, Video Understanding, Computer Vision, Visual Question Answering, Model Evaluation
Machine Learning	PyTorch, Hugging Face Transformers, PyTorch Geometric, scikit-learn
ML Systems	SGLang, ONNX Runtime, FastAPI, Docker
Programming & Data	Python, SQL, PostgreSQL, pgvector
Language	English — TOEIC Listening & Reading 930/990